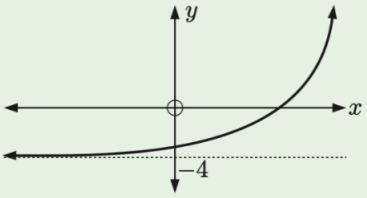
EXERCISE 1

[12 marks]

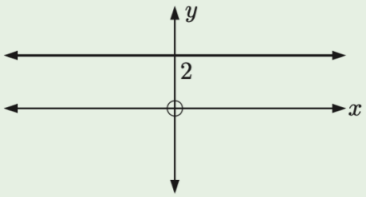
1 For each graph, state:

i the domain **ii** the range **iii** whether the graph shows a function.

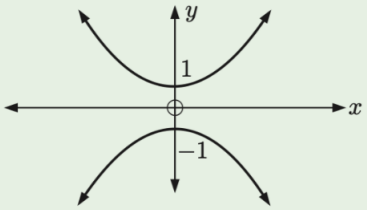
a



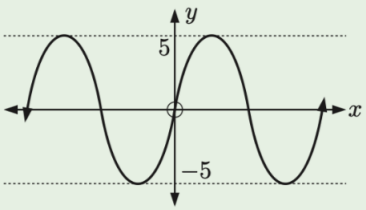
b



c



d



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EXERCISE 2 [4 marks]

State the value(s) of x for which $f(x)$ is undefined :

a. $f(x) = 10 + \dfrac{3}{2x-1}$

b. $f(x) = \sqrt{x+7}$

EXERCISE 3 [6 marks]

If $F(x) = 2x^2 + 3x - 1$, find in simplest form :

1. $F(-x)$

2. $F(x+h)$

EXERCISE 4

1. Given $f(x) = ax + \frac{b}{x}$, $f(1) = 1$, and $f(2) = 5$, find constants a and b. [4 marks]

2. [BONUS] Given $T(x) = ax^2 + bx + c$, $T(0) = -4$, $T(1) = -2$, and $T(2) = 6$, find a, b, and c. [2 marks]